

Question ID e7b6f0d1

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	

ID: e7b6f0d1

SAT2 M 1.1

$$4x + 6 = 18$$

Which equation has the same solution as the given equation?

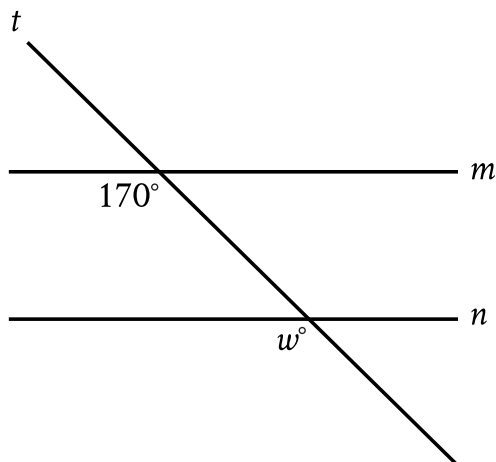
- A. $4x = 108$
- B. $4x = 24$
- C. $4x = 12$
- D. $4x = 3$

Question ID 5207e508

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 5207e508

SAT2 M 1.2



Note: Figure not drawn to scale.

In the figure, line m is parallel to line n . What is the value of w ?

- A. 17
- B. 30
- C. 70
- D. 170

Question ID 7760c516

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: 7760c516

SAT2 M 1.3

Each value in the data set shown represents the height, in centimeters, of a plant.

6, 10, 13, 2, 15, 22, 10, 4, 4, 4

What is the mean height, in centimeters, of these plants?

Question ID 637022d2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	

ID: 637022d2

SAT2 M 1.4

$2.5b + 5r = 80$

The given equation describes the relationship between the number of birds, b , and the number of reptiles, r , that can be cared for at a pet care business on a given day. If the business cares for **16** reptiles on a given day, how many birds can it care for on this day?

- A. 0
- B. 5
- C. 40
- D. 80

Question ID e336a1d2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: e336a1d2

SAT2 M 1.5

A cube has an edge length of **41** inches. What is the volume, in cubic inches, of the cube?

- A. **164**
- B. **1,681**
- C. **10,086**
- D. **68,921**

Question ID e9f4521a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	

ID: e9f4521a

SAT2 M 1.6

13 is $p\%$ of 25. What is the value of p ?

Question ID c048055c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: c048055c

SAT2 M 1.7

A model predicts that the population of Springfield was **15,000** in **2005**. The model also predicts that each year for the next **5** years, the population p increased by **4%** of the previous year's population. Which equation best represents this model, where x is the number of years after **2005**, for $x \leq 5$?

- A. $p = 0.96(15,000)^x$
- B. $p = 1.04(15,000)^x$
- C. $p = 15,000(0.96)^x$
- D. $p = 15,000(1.04)^x$

Question ID 13e57f0a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 13e57f0a

SAT2 M 1.8

$$-4x^2 - 7x = -36$$

What is the positive solution to the given equation?

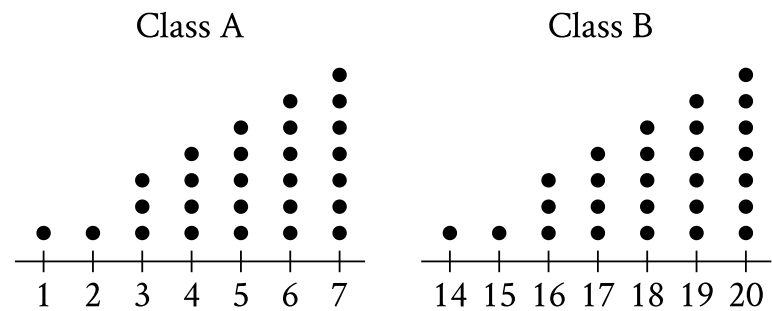
- A. $\frac{7}{4}$
- B. $\frac{9}{4}$
- C. 4
- D. 7

Question ID d94018fd

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: d94018fd

SAT2 M 1.9



Each of the dot plots shown represents the number of glue sticks brought in by each student for two classes, class A and class B. Which statement best compares the standard deviations of the numbers of glue sticks brought in by each student for these two classes?

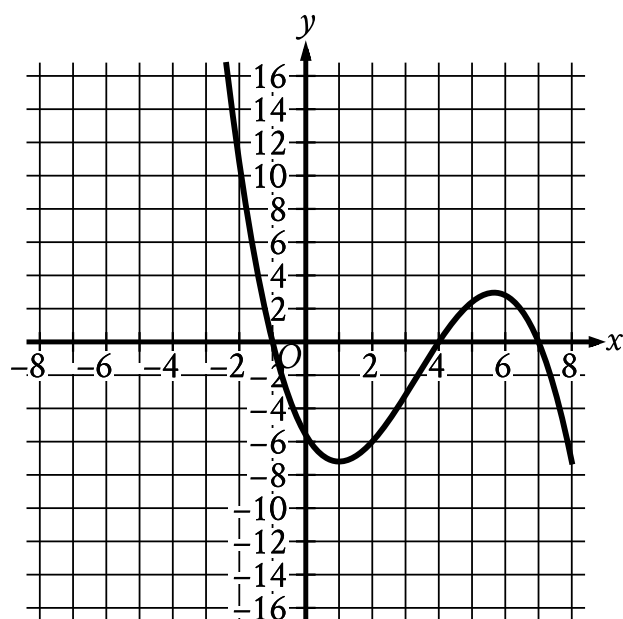
- A. The standard deviation of the number of glue sticks brought in by each student for class A is less than the standard deviation of the number of glue sticks brought in by each student for class B.
- B. The standard deviation of the number of glue sticks brought in by each student for class A is equal to the standard deviation of the number of glue sticks brought in by each student for class B.
- C. The standard deviation of the number of glue sticks brought in by each student for class A is greater than the standard deviation of the number of glue sticks brought in by each student for class B.
- D. There is not enough information to compare these standard deviations.

Question ID cc6ccd71

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: cc6ccd71

SAT2 M 1.10



The graph of $y = f(x)$ is shown, where the function f is defined by $f(x) = ax^3 + bx^2 + cx + d$ and a , b , c , and d are constants. For how many values of x does $f(x) = 0$?

- A. One
- B. Two
- C. Three
- D. Four

Question ID f1c81b3b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: f1c81b3b

SAT2 M 1.11

The exponential function g is defined by $g(x) = 19 \cdot a^x$, where a is a positive constant. If $g(3) = 2,375$, what is the value of $g(4)$?

Question ID 895628b5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 895628b5

SAT2 M 1.12

$$y = (x - 2)(x + 4)$$
$$y = 6x - 12$$

Which ordered pair (x, y) is the solution to the given system of equations?

- A. $(0, 2)$
- B. $(-4, 2)$
- C. $(2, 0)$
- D. $(2, -4)$

Question ID bcb66188

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: bcb66188

SAT2 M 1.13

Triangle FGH is similar to triangle JKL , where angle F corresponds to angle J and angles G and K are right angles. If $\sin(F) = \frac{308}{317}$, what is the value of $\sin(J)$?

- A. $\frac{75}{317}$
- B. $\frac{308}{317}$
- C. $\frac{317}{308}$
- D. $\frac{317}{75}$

Question ID 90bcaa61

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 90bcaa61

SAT2 M 1.14

The function $f(t) = 60,000(2)^{\frac{t}{410}}$ gives the number of bacteria in a population t minutes after an initial observation. How much time, in minutes, does it take for the number of bacteria in the population to double?

Question ID 95ed0b69

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 95ed0b69

SAT2 M 1.15

$$p = \frac{k}{4j+9}$$

The given equation relates the distinct positive numbers p , k , and j . Which equation correctly expresses $4j + 9$ in terms of p and k ?

- A. $4j + 9 = \frac{k}{p}$
- B. $4j + 9 = kp$
- C. $4j + 9 = k - p$
- D. $4j + 9 = \frac{p}{k}$

Question ID db422e7f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	■ ■ ■

ID: db422e7f

SAT2 M 1.16

Line p is defined by $4y + 8x = 6$. Line r is perpendicular to line p in the xy -plane. What is the slope of line r ?

Question ID 249d3f80

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: 249d3f80

SAT2 M 1.17

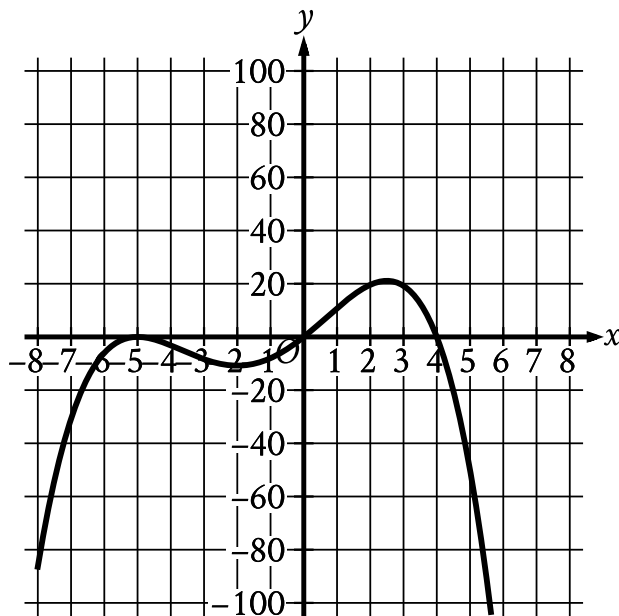
Point O is the center of a circle. The measure of arc RS on this circle is 100° . What is the measure, in degrees, of its associated angle ROS ?

Question ID 252a3b3a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 252a3b3a

SAT2 M 1.18



Which of the following could be the equation of the graph shown in the xy -plane?

- A. $y = -\frac{1}{10}x(x-4)(x+5)$
- B. $y = -\frac{1}{10}x(x-4)(x+5)^2$
- C. $y = -\frac{1}{10}x(x-5)(x+4)$
- D. $y = -\frac{1}{10}x(x-5)^2(x+4)$

Question ID 9b5b23fc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	■ ■ ■

ID: 9b5b23fc

SAT2 M 1.19

For $x > 0$, the function f is defined as follows:

$f(x)$ equals 201% of x

Which of the following could describe this function?

- A. Decreasing exponential
- B. Decreasing linear
- C. Increasing exponential
- D. Increasing linear

Question ID 841ef26c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 841ef26c

SAT2 M 1.20

$$f(x) = 4x^2 + 64x + 262$$

The function g is defined by $g(x) = f(x + 5)$. For what value of x does $g(x)$ reach its minimum?

- A. -13
- B. -8
- C. -5
- D. -3

Question ID 916ffe9b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Inference from sample statistics and margin of error	■ ■ ■

ID: 916ffe9b

SAT2 M 1.21

Poll Results	
Angel Cruz	483
Terry Smith	320

The table shows the results of a poll. A total of **803** voters selected at random were asked which candidate they would vote for in the upcoming election. According to the poll, if **6,424** people vote in the election, by how many votes would Angel Cruz be expected to win?

- A. **163**
- B. **1,304**
- C. **3,864**
- D. **5,621**

Question ID 77c0cced

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	■ ■ ■

ID: 77c0cced

SAT2 M 1.22

$$y = 2x^2 - 21x + 64$$
$$y = 3x + a$$

In the given system of equations, a is a constant. The graphs of the equations in the given system intersect at exactly one point, (x, y) , in the xy -plane. What is the value of x ?

- A. -8
- B. -6
- C. 6
- D. 8